

Q12..PROGRAM TO PRINT ASCII CODE OF ANY CHARACTER.

```
C 12.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5
6      printf("Enter any character :");
7      scanf("%c",&ch);
8
9      printf("The ASCII value of %c is %d \n",ch,ch);
10 }
```

Problems Output Debug Console Terminal Ports

- Enter any character :W
The ASCII value of W is 87
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 12.c -o 12 && "/Users/yagnapatel/Documents/IOP.c/"12
Enter any character :a
The ASCII value of a is 97
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 12.c -o 12 && "/Users/yagnapatel/Documents/IOP.c/"12
Enter any character :4
The ASCII value of 4 is 52
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 12.c -o 12 && "/Users/yagnapatel/Documents/IOP.c/"12
Enter any character :*
The ASCII value of * is 42

Code Code Code

Q13. PROGRAM TO CHECK WHETHER A NO. IS EVEN OR ODD.

```
C 13.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int i;
5      printf("Enter any value :");
6      scanf("%d",&i);
7
8
9      if(i % 2 == 0)
10     {
11         printf("The no. %d is even\n",i);
12     }
13     else {
14         printf("The no. %d is odd\n",i);
15     }
16 }
```

Problems Output Debug Console **Terminal** Ports

- Enter any value :4
The no. 4 is even
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 13.c -o 13 && "/Users/yagnapatel/Documents/IOP.c/"13
Enter any value :5
The no. 5 is odd
- yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q14. PROGRAM TO CHECK WHETHER A GIVEN CHARACTER IS CAPITAL, LETTER, SMALL CASE LETTER, A DIGIT OR A SPECIAL SYMBOL.

```
C 14.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      char ch;
6
7      printf("enter any character: ");
8      scanf("%c", &ch);
9
10     if (ch >= 'A' && ch <= 'Z')
11     {
12         printf("capital letter\n");
13     }
14     else if (ch >= 'a' && ch <= 'z')
15     {
16         printf("small letter\n");
17     }
18     else if (ch >= '0' && ch <= '9')
19     {
20         printf("digit\n");
21     }
22     else
23     {
24         printf("special symbol\n");
25     }
26 }
```

Problems Output Debug Console **Terminal** Ports

```
• enter any character: H
capital letter
○ yagnapatel@Yagnas-MacBook-Air IOP.c %
```

Code Code Code

Q15. PROGRAM TO READ MARKS OF FIVE SUBJECT AND PRINT DIVISION.

```
C 15.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int s1,s2,s3,s4,s5,percentage;
5      printf("Enter marks of five subject(out of 100) : ");
6      scanf("%d%d%d%d%d",&s1,&s2,&s3,&s4,&s5);
7
8      percentage =(s1+s2+s3+s4+s5)/5;
9      printf("The total maarks of five subject is %d \n",(s1+s2+s3+s4+s5));
10     printf("The percentage is %d \n",percentage);
11
12     if(percentge>=80){
13         printf("First division\n");
14     }
15     else if(percentge>=65){
16         printf("Second division\n");
17     }
18     else if(percentge>=50){
19         printf("Third division\n");
20     }
21     else{
22         printf("Fail\n");
23     }
24 }
```

Problems 3 Output Debug Console Terminal Ports

```
Enter marks of five subject(out of 100) : 54
67
97
97
68
The total maarks of five subject is 383
The percentage is 76
Second division
yagnapate@Yagnas-MacBook-Air IOP.c %
```

Code
Code
Code

Q16.PROGRAM TO READ THREE NOS. AND PRINT MAX.

```
C 16.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int a,b,c;
5      printf("Enter three integers :");
6      scanf("%d%d%d",&a,&b,&c);
7
8      if(a>=b)
9      {
10         if(a>=c)
11         {
12             printf("maximum is %d\n",a);
13         }
14         else
15         {
16             printf("maximum is %d\n",c);
17         }
18     }
19     else
20     {
21         if(b>=c)
22         {
23             printf("maximum is %d\n",b);
24         }
25         else
26         {
27             printf("maximum is %d\n",c);
28         }
29     }
30 }
31
```

Problems 3 Output Debug Console Terminal Ports

● Enter three integers :4
7
45
maximum is 45
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q17. PROGRAM TO READ THREE NOS. AND PRINT MAX USING LOGICAL OPERATOR.(&&)

```
C 17.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int a, b, c;
5
6      printf("Enter three integers: ");
7      scanf("%d%d%d", &a, &b, &c);
8
9      if (a >= b && a >= c)
10     {
11         printf("Maximum is: %d\n", a);
12     }
13     else if (b >= a && b >= c)
14     {
15         printf("Maximum is: %d\n", b);
16     }
17     else
18     {
19         printf("Maximum is: %d\n", c);
20     }
21 }
```

Problems 3 Output Debug Console Terminal Ports

Enter three integers: 64
53
75
Maximum is: 75
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q18. PROGRAM TO READ THREE NOS. AND PRINT MAX USING CONDITIONAL (? :) OPERATOR.

```
C 18.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int a, b, c, max;
5
6      printf("Enter three integers: ");
7      scanf("%d %d %d", &a, &b, &c);
8
9      max = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);
10
11     printf("Maximum is: %d\n", max);
12 }
```

Problems 3 Output Debug Console Terminal Ports

```
Enter three integers: 53
64
24
Maximum is: 64
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

Code Code Code

Q19.PROGRAM TO READ A CHARACTER AND CHECK WHETHER IT IS A SMALL CASE LETTER OR NOT USING CONDONAL OPERATOR.

```
C 19.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5
6      printf("enter a character: ");
7      scanf("%c", &ch);
8
9      (ch >= 'a' && ch <= 'z') ? printf("it is small letter\n") : printf("not a small letter\n");
10 }
```

Problems 3 Output Debug Console Terminal Ports

- enter a character: E
not a small letter
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 19.c -o 19 && "/Users/yagnapatel/Documents/IOP.c/"19
enter a character: r
it is small letter
- yagnapatel@Yagnas-MacBook-Air IOP.c %

Code Code Code

Q20. PROGRAM WHICH WILL WORK LIKE A SIMPLE CALCULATOR USING SWITCH-CASE.

```
C 20.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5      int a,b;
6
7      printf("Enter the character (+,-,*,/): ");
8      scanf("%c",&ch);
9
10     printf("Enter two numbers : ");
11     scanf("%d%d",&a,&b);
12
13     switch(ch)
14     {
15     case '+':
16         printf("Addition of %d and %d is : %d",a,b,a+b);
17         break;
18     case '-':
19         printf("Subtraction of %d and %d is : %d",a,b,a-b);
20         break;
21     case '*':
22         printf("Multiplication of %d and %d is : %d",a,b,a*b);
23         break;
24     case '/':
25         if(b!=0)
26             printf("Division of %d and %d is : %d",a,b,a/b);
27         else
28             printf("Division by zero is not allowed.");
29     }
30 }
```

Problems 3 Output Debug Console Terminal Ports

- Enter the character (+,-,*,/): +
Enter two numbers : 56
65
Addition of 56 and 65 is : 121
- yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 20.c -o 20 && "/Users/yagnapatel/Documents/IOP.c/"20
Enter the character (+,-,*,/): *
Enter two numbers : 65
75
Multiplication of 65 and 75 is : 4875
- yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q21. PROGRAM TO READ TWO NOS. AND PERFORM SPECIFIC TASK (USING ARITHMETIC OPERATOR) AND THIS PERFORM USING SWITCH-CASE. (CASE IN CHAR FORM).

```
C 21.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      char ch;
5      int a,b;
6
7      printf("Enter the character (+,-,*,/): ");
8      scanf("%c",&ch);
9
10     printf("Enter two numbers : ");
11     scanf("%d%d",&a,&b);
12
13     switch(ch)
14     {
15     case '+':
16         printf("Addition of %d and %d is : %d",a,b,a+b);
17         break;
18     case '-':
19         printf("Subtraction of %d and %d is : %d",a,b,a-b);
20         break;
21     case '*':
22         printf("Multiplication of %d and %d is : %d",a,b,a*b);
23         break;
24     case '/':
25         if(b!=0)
26             printf("Division of %d and %d is : %d",a,b,a/b);
27         else
28             printf("Division by zero is not allowed.");
29     }
30 }
```

Problems 3 Output Debug Console Terminal Ports

Enter the character (+,-,*,/): -
Enter two numbers : 54
34
Subtraction of 54 and 34 is : 20
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q22.PROGRAM TO CALCULATE FACTORIAL OF A NO

```
C 22.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int i,n,f=1;
5      printf("Enter the value for factorial :");
6      scanf("%d",&n);
7
8      for(i=1;i<=n;i++)
9      {
10         f = f * i;
11     }
12     printf("Factorial = %d\n",f);
13 }
14
```

Problems 9 Output Debug Console Terminal Ports

```
Enter the value for factorial :9
Factorial = 362880
yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 22.c -o 22 && "/Users/yag
napatel/Documents/IOP.c/"22
Enter the value for factorial : 6
Factorial = 720
yagnapatel@Yagnas-MacBook-Air IOP.c %
```

Code Code Code

Q23. PROGRAM TO READ TWO NOS. AND CALCULATE POWER WITHOUT USING HEADER FILE(<MATH.H>).

```
C 23.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int base, exp, i, result = 1;
5
6      printf("Enter base: ");
7      scanf("%d", &base);
8      printf("Enter exponent: ");
9      scanf("%d", &exp);
10
11     for (i = 1; i <= exp; i++) {
12         result = result * base;
13     }
14     printf("Result = %d\n", result);
15 }
```

Problems 3 Output Debug Console Terminal Ports

Enter base: 4
Enter exponent: 6
Result = 4096

yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q24. WRITE A C PROGRAM TO FIND THE VALUE OF Y USING

$Y(x,n) = 1+x$ where $n=1$

$Y(x,n) = 1+x/n$ where $n=2$

$Y(x,n) = 1 + x^n$ when $n=3$

$Y(x,n) = 1 + nx$ when $n>3$ or $n<1$

```
C 24.c > main()
1  #include <stdio.h>
2  int main()
3  {
4      int x, n;
5      float y;
6      printf("Enter n and x: ");
7      scanf("%d %d", &n, &x);
8
9      if (n == 1) {
10         y = 1 + x;
11     } else if (n == 2) {
12         y = 1.0 + ((float)x / 2);
13     } else if (n == 3) {
14         y = 1 + (x * x * x);
15     } else {
16         y = 1 + (n * x);
17     }
18     printf("Y = %.2f\n", y);
19 }
```

Problems 3 Output Debug Console Terminal Ports

Enter n and x: 3
54
Y = 157465.00
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q25.PROGRAM TO PRINT TABLE OF ANY NO.

```
C 25.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int i,n;
5      printf("Enter a number to find any table for a number :");
6      scanf("%d",&n);
7
8      for(i=1;i<=10;i++)
9      {
10         printf("%d * %d = %d\n",n,i, i * n);
11     }
12 }
```

Problems 3 Output Debug Console Terminal Ports

yagnapatel@Yagnas-MacBook-Air IOP.c % cd "/Users/yagnapatel/Documents/IOP.c/" && gcc 25.c -o 25 && "/Users/yag
napatel/Documents/IOP.c/"25
Enter a number to find any table for a number :89
89 * 1 = 89
89 * 2 = 178
89 * 3 = 267
89 * 4 = 356
89 * 5 = 445
89 * 6 = 534
89 * 7 = 623
89 * 8 = 712
89 * 9 = 801
89 * 10 = 890
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code

Q26. PROGRAM TO PRINT SUM OF ALL ODD AND EVEN NO B/W 1 TO N.

```
C 26.c > main()
1  #include<stdio.h>
2  int main()
3  {
4      int even=0,odd=0,n,i;
5      printf("Enter n :");
6      scanf("%d",&n);
7
8      for(i=1;i<=n;i++)
9      {
10         if(i % 2 == 0){
11             even = even + i;}
12         else{
13             odd = odd + i;
14         }
15     }
16     printf("Sum of even numbers is : %d\n",even);
17     printf("Sum of odd numbers is : %d\n",odd);
18 }
```

Problems 3 Output Debug Console Terminal Ports

Enter n :6
Sum of even numbers is : 12
Sum of odd numbers is : 9
yagnapatel@Yagnas-MacBook-Air IOP.c %

Code
Code
Code